SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Engineering Thermodynamics: Laws, Rankine & Refrigeration Cycles

Department: MECHANICAL	**Course / Subject:** Engineering Thermodynamics (ME301PC)
Semester: Semester 3	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Zeroth, First & Second Laws of Thermodynamics, Entropy generation, Pure substance Phase diagrams, Ideal Gas laws, Rankine steam cycles, and Vapour Compression.

Unit 1: Basic Concepts & First Law

System, surroundings, boundaries, Thermodynamic equilibrium, State postulate, Quasi-static process, Work and heat interactions, First Law for closed/open systems (SFEE).

Unit 2: Second Law & Entropy

Kelvin-Planck and Clausius statements, Reversibility, Carnot cycle efficiency, Clausius inequality, Entropy principle, T-s diagrams, Available and unavailable energy (Exergy).

Unit 3: Properties of Pure Substances

P-V-T surfaces, p-v, T-s, h-s (Mollier) diagrams for water, Dryness fraction, Steam tables usage, Throttling and Separating calorimeters.

Unit 4: Thermodynamic Power Cycles

Air standard cycles: Otto cycle, Diesel cycle, Dual combustion cycle, Thermal efficiency comparisons, Mean Effective Pressure (MEP) derivations.

Unit 5: Steam Cycles & Refrigeration

Simple Rankine cycle, Reheat cycle, Regenerative feedwater heating cycle, Vapor Compression Refrigeration System (VCRS), COP calculation, Eco-friendly refrigerants.